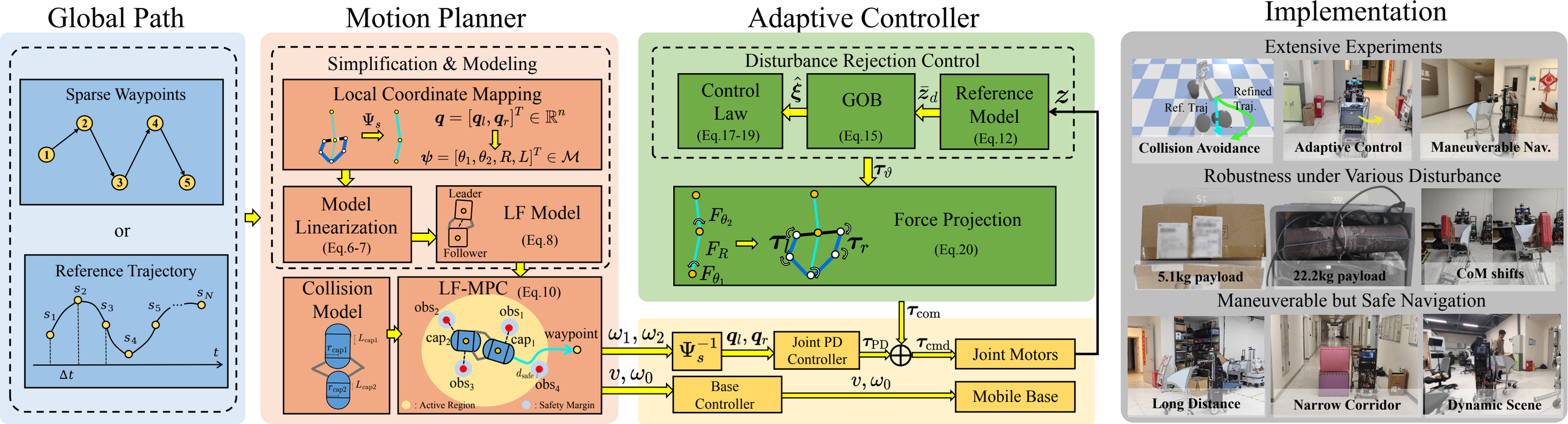

Global Path
Sparse Waypoints
1
2
3
4
5
or
Reference Trajectory
s_1
s_2
s_3
s_4
s_5
s_N
t
Δt
Motion Planner
Simplification & Modeling
Local Coordinate Mapping
Ψ_s
$q = [q_l, q_r]^T \in \mathbb{R}^n$
$\psi = [\theta_1, \theta_2, R, L]^T \in \mathcal{M}$
Model Linearization
(Eq.6-7)
Leader
Follower
LF Model
(Eq.8)
Collision Model
r_{cap1}
L_{cap1}
r_{cap2}
L_{cap2}
LF-MPC (Eq.10)
obs_1
obs_2
obs_3
obs_4
cap_1
cap_2
waypoint
d_{safe}
: Active Region
: Safety Margin
Adaptive Controller
Disturbance Rejection Control
Control Law
(Eq.17-19)
$\hat{\xi}$
GOB
(Eq.15)
$\bar{z}_d$
Reference Model
(Eq.12)
z
τ_ϑ
F_{θ_2}
F_R
F_{θ_1}
τ_l
τ_r
Force Projection
(Eq.20)
τ_{com}
ω_1, ω_2
Ψ_s^{-1}
q_l, q_r
Joint PD Controller
τ_{PD}
τ_{cmd}
Joint Motors
v, ω_0
Base Controller
v, ω_0
Mobile Base
Implementation
Extensive Experiments
Ref. Traj
Refined Traj.
Collision Avoidance
Adaptive Control
Maneuverable Nav.
Robustness under Various Disturbance
5.1kg payload
22.2kg payload
CoM shifts
Maneuverable but Safe Navigation
Long Distance
Narrow Corridor
Dynamic Scene